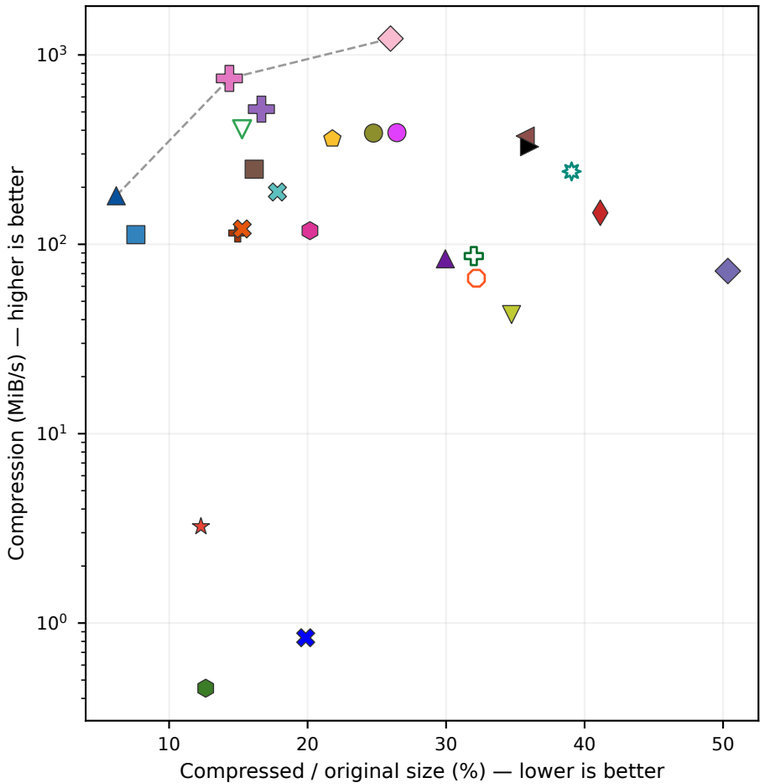
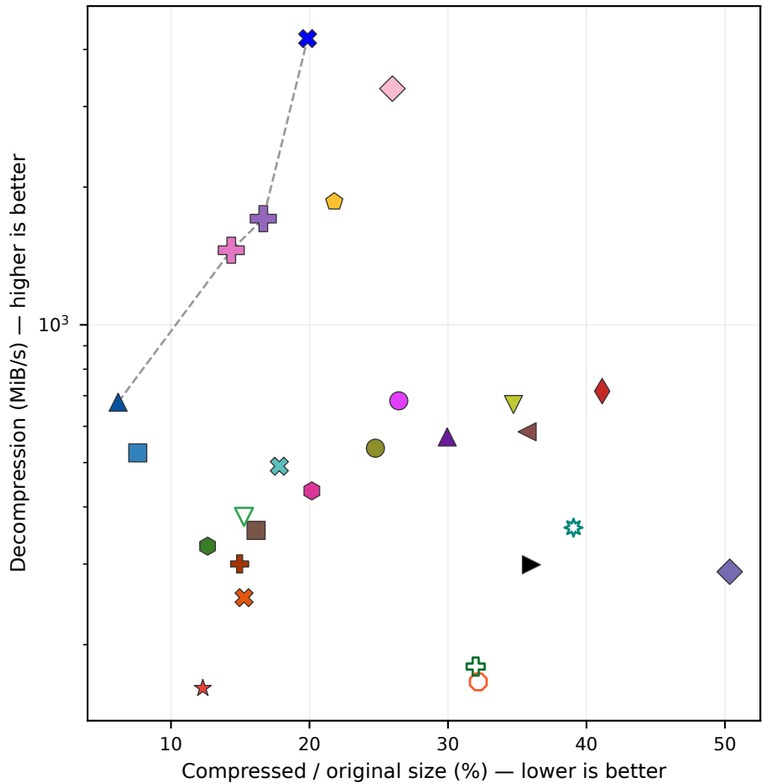
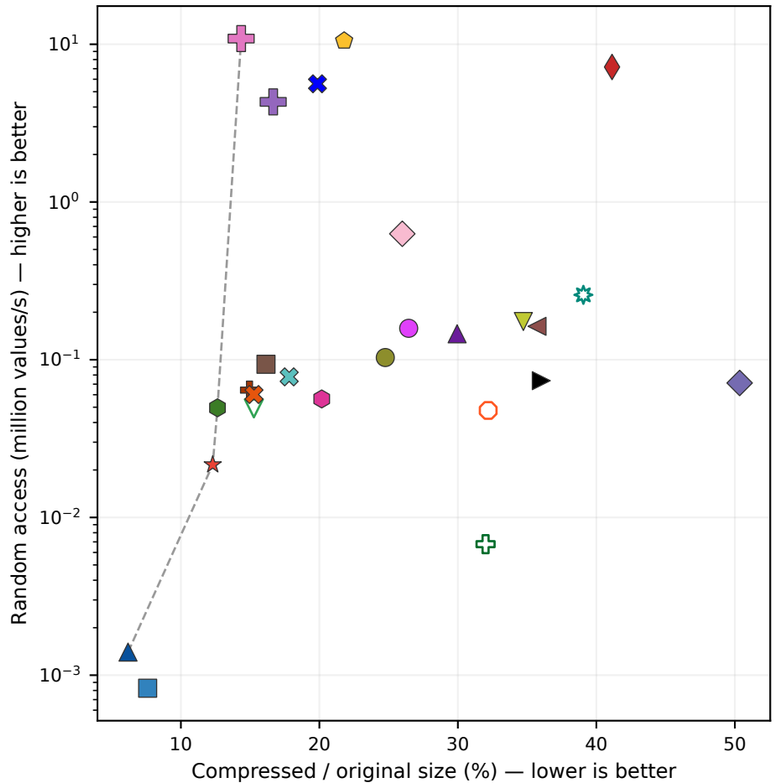
AP • optimized GEF 07d4e4a (uniform K=32,000) vs competitors



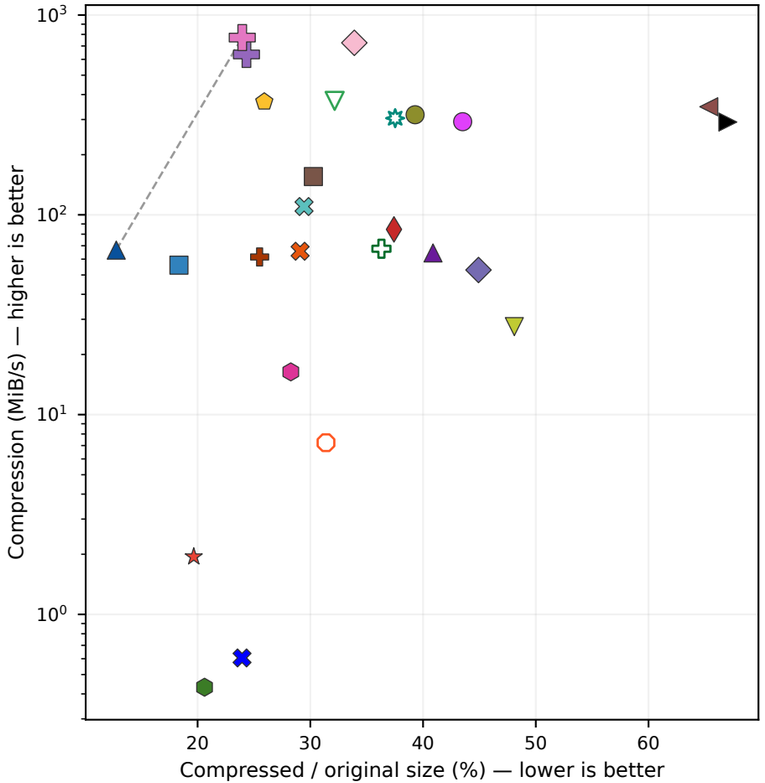


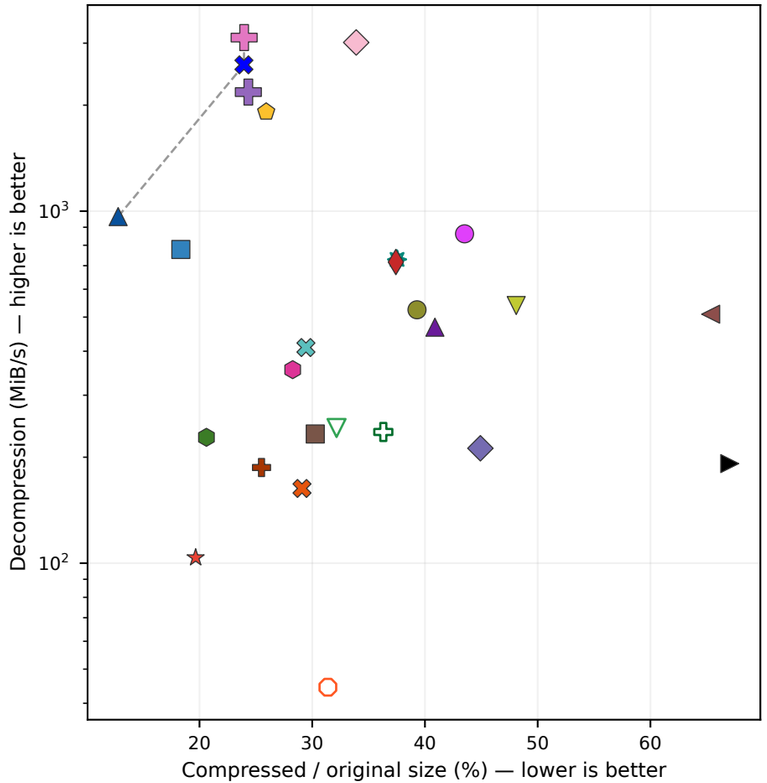


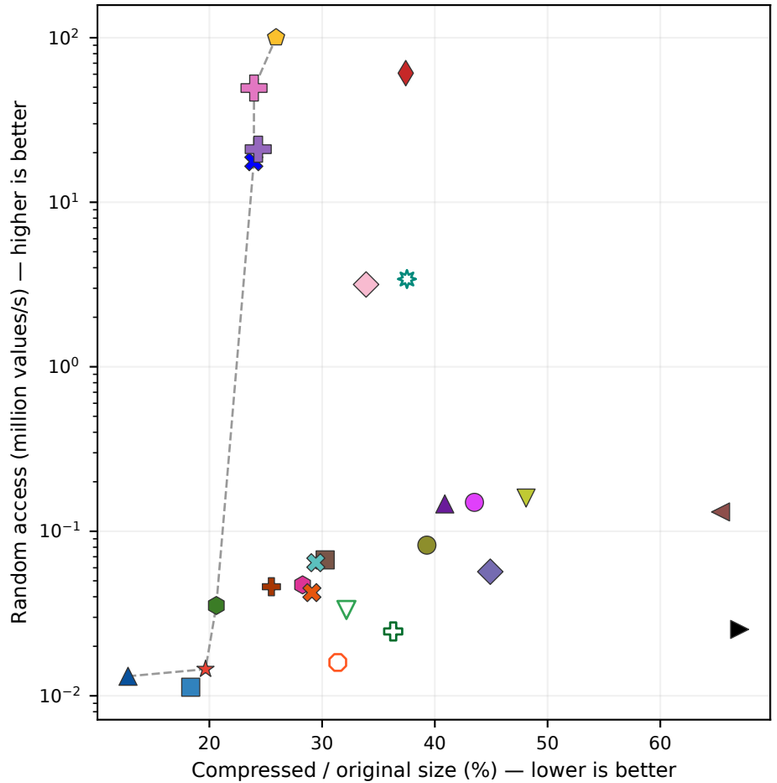
ALP
B*-GEF approx. [07d4e4a]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [07d4e4a]
SElf*
Snappy
TSXor
Xz
Zstd

GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

BM • optimized GEF 07d4e4a (uniform K=32,000) vs competitors



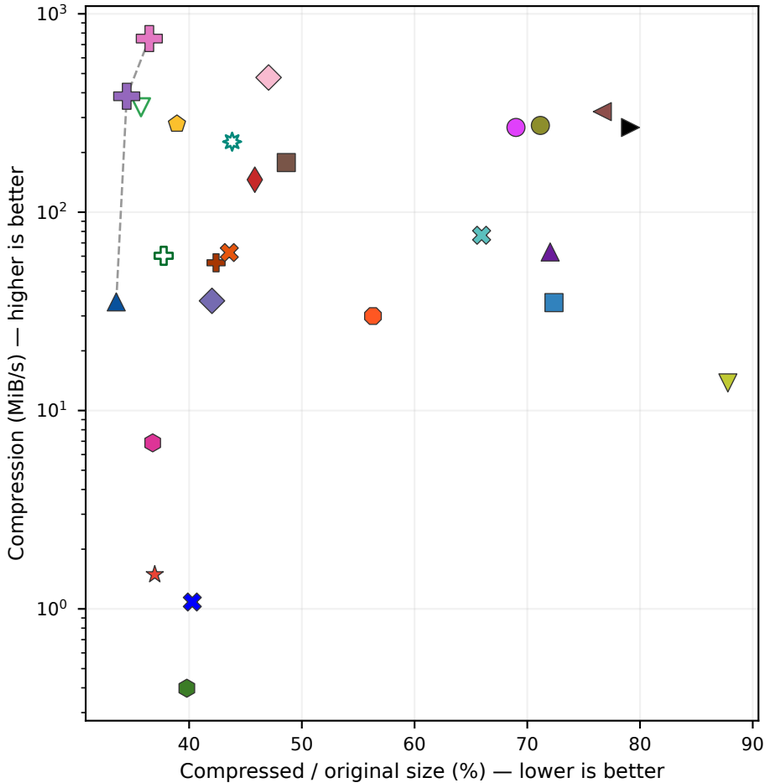


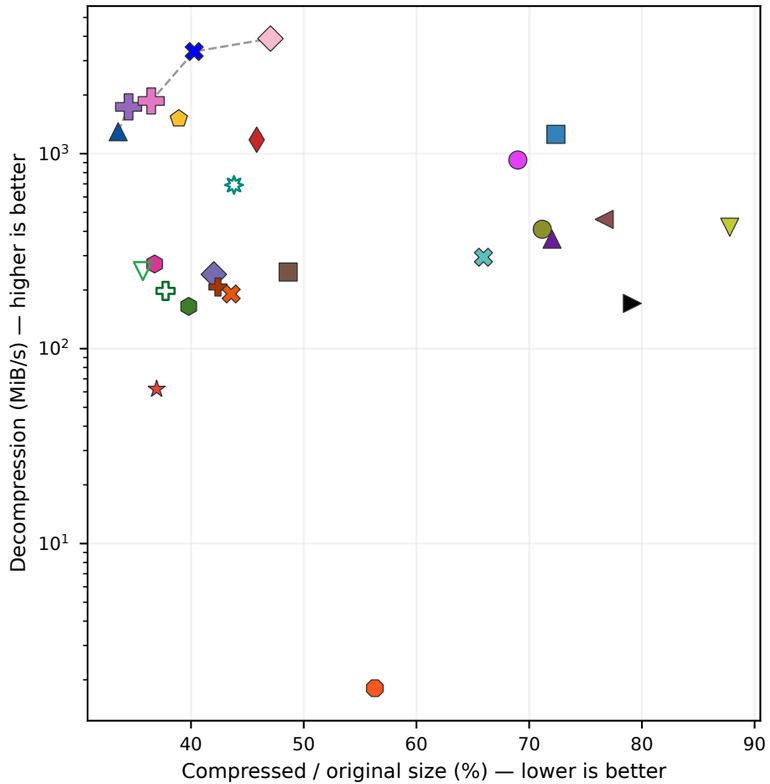


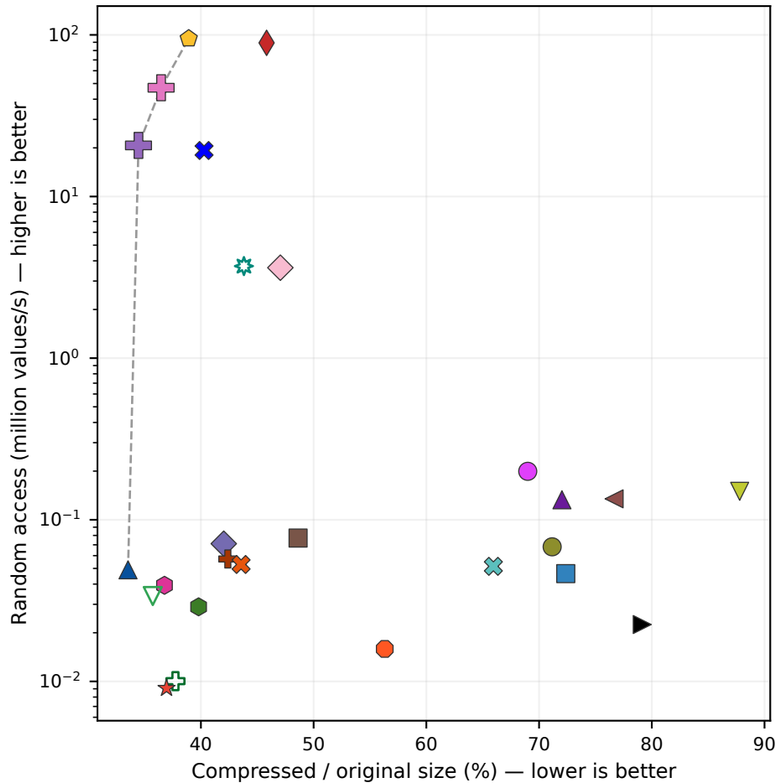
ALP
B*-GEF approx. [07d4e4a]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [07d4e4a]
SElf*
Snappy
TSXor
Xz
Zstd

GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

BP • optimized GEF 07d4e4a (uniform K=32,000) vs competitors



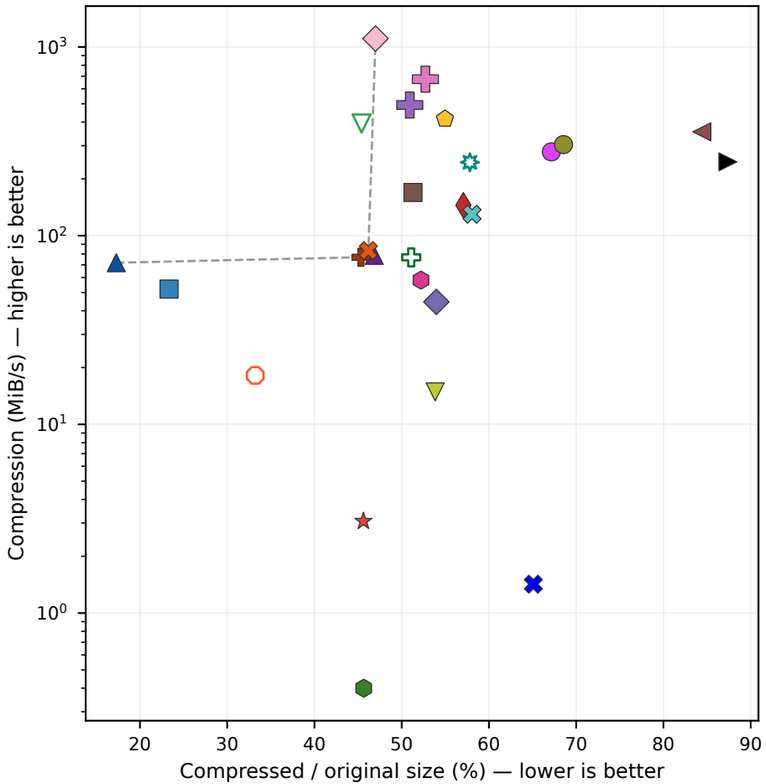


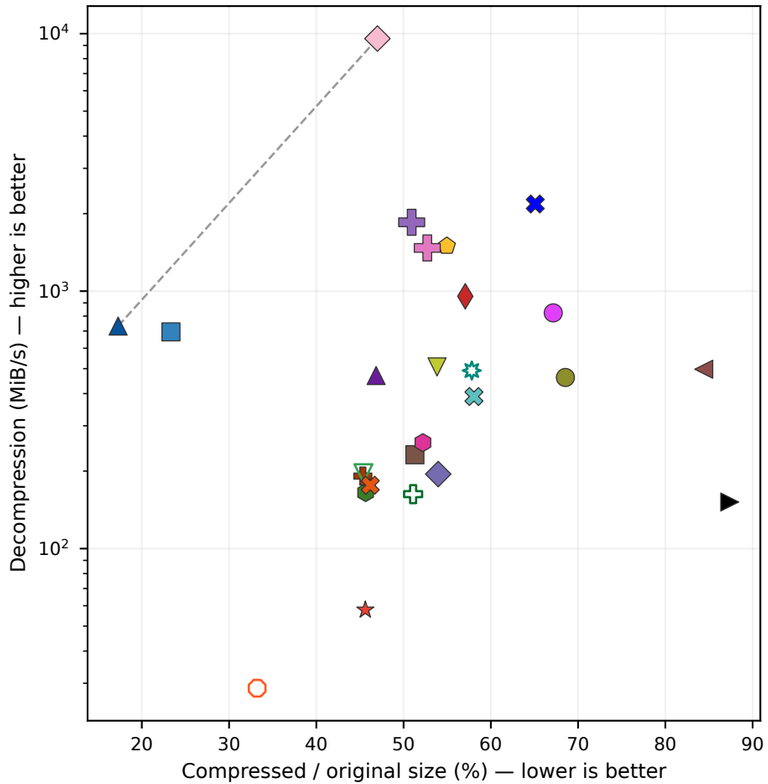


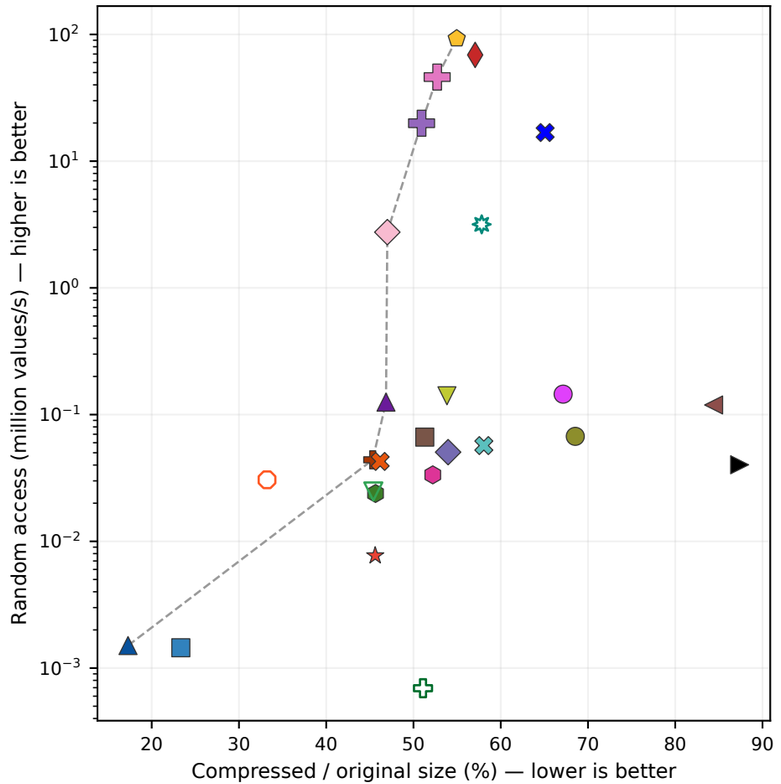
ALP
B*-GEF approx. [07d4e4a]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [07d4e4a]
SElf*
Snappy
TSXor
Xz
Zstd

GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

BT • optimized GEF 07d4e4a (uniform K=32,000) vs competitors

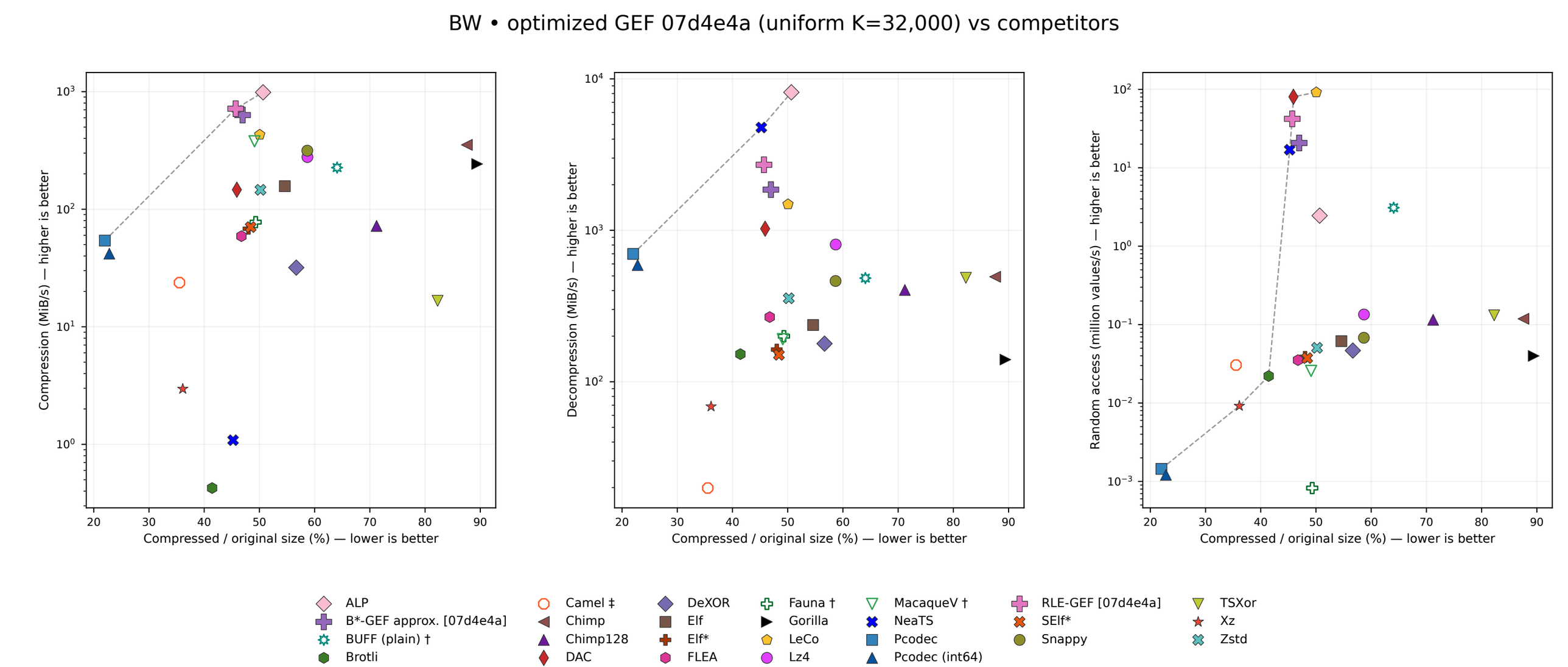




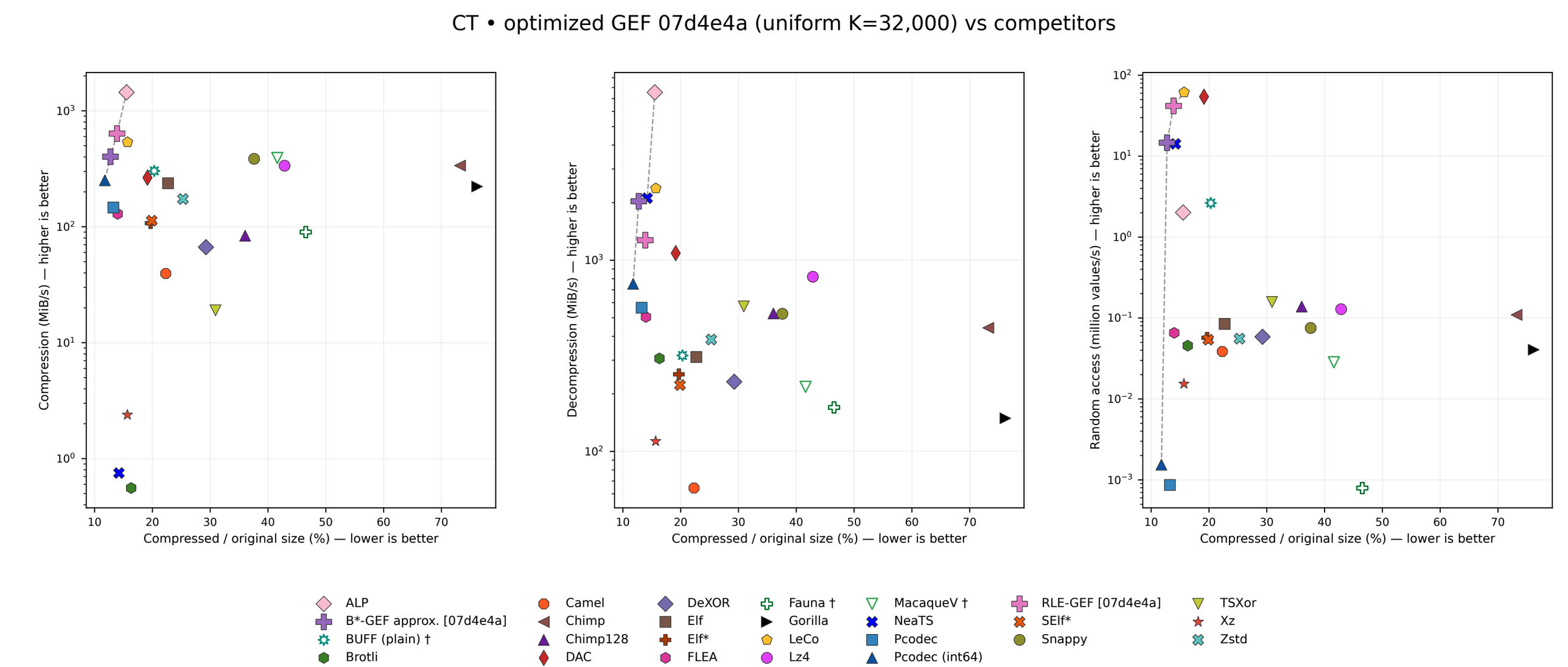


ALP
B*-GEF approx. [07d4e4a]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [07d4e4a]
SElf*
Snappy
TSXor
Xz
Zstd

GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

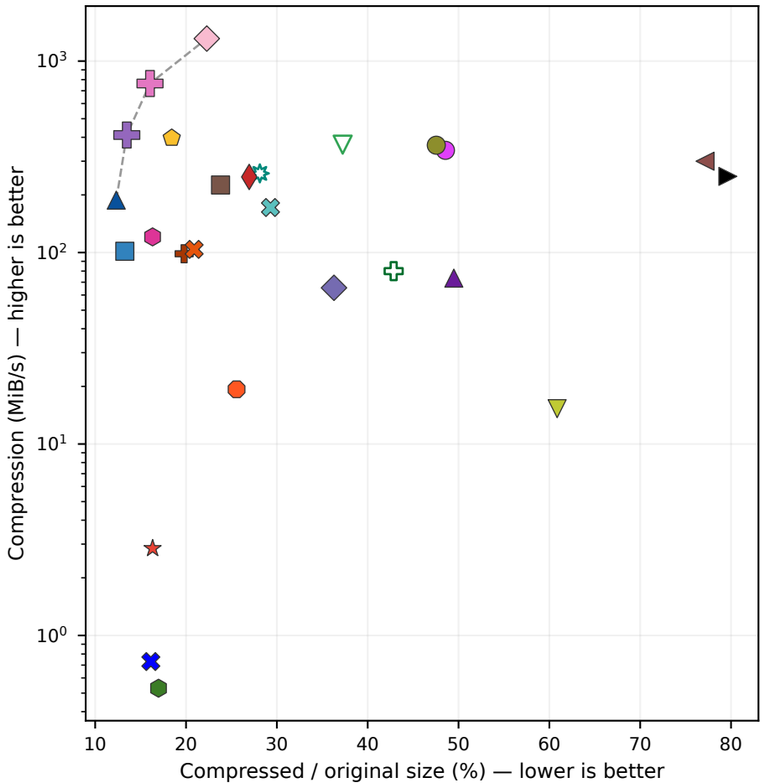


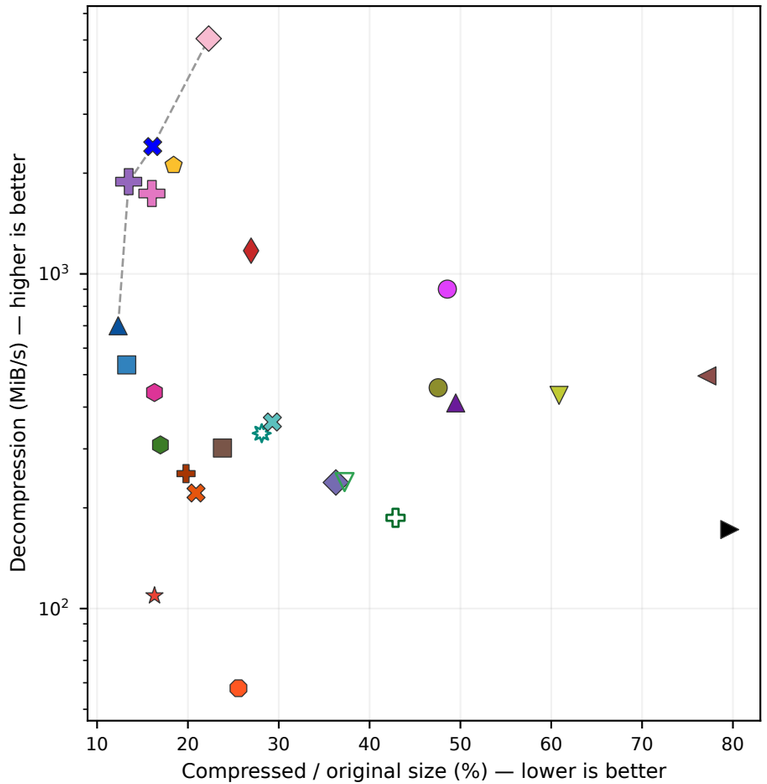
GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

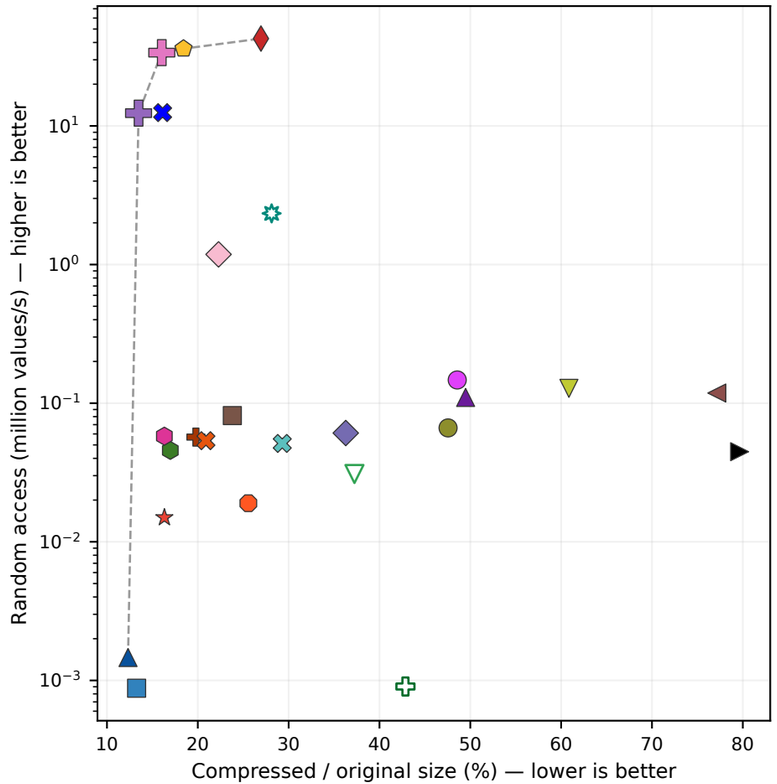


GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

DP • optimized GEF 07d4e4a (uniform K=32,000) vs competitors



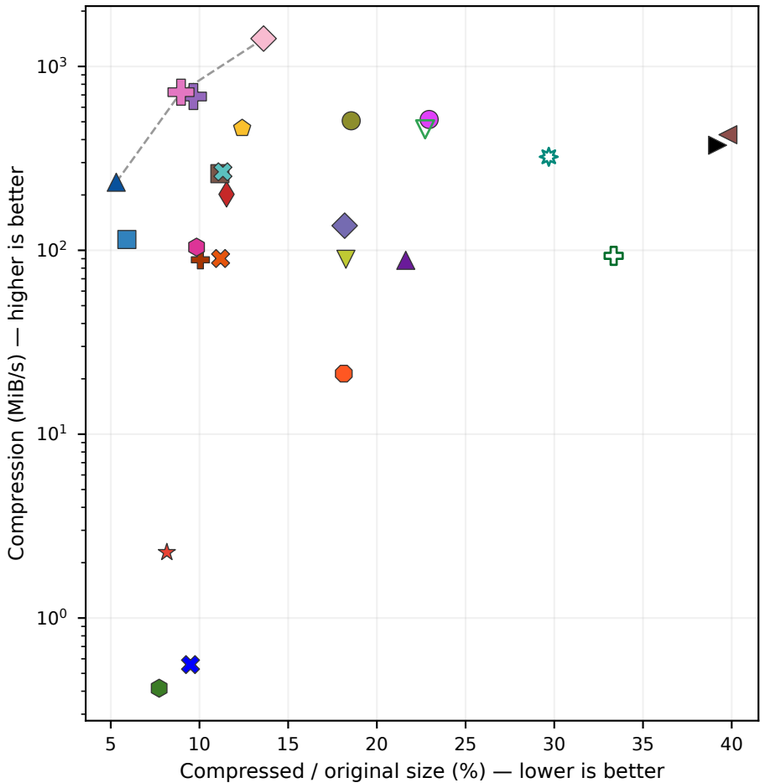


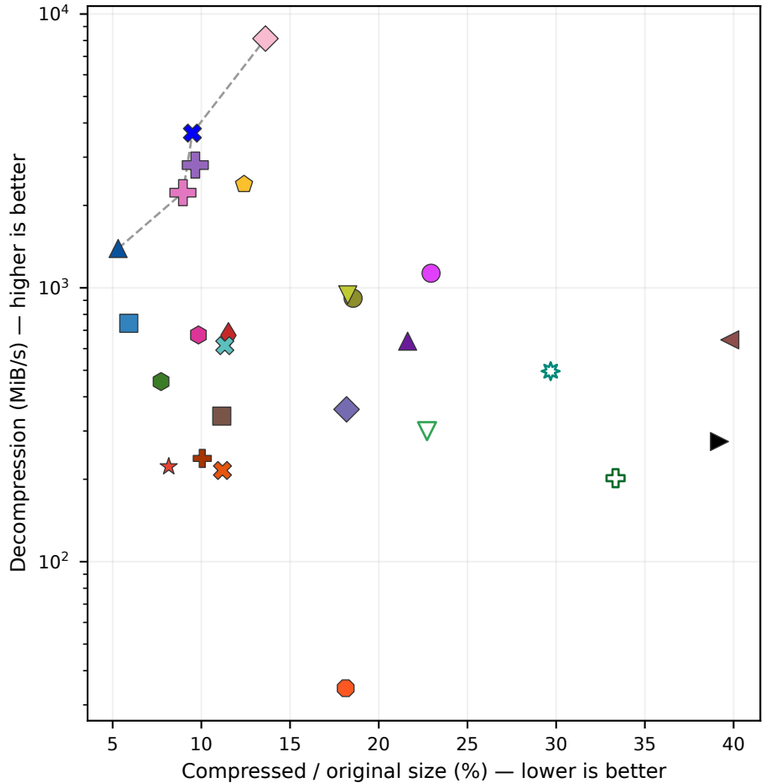


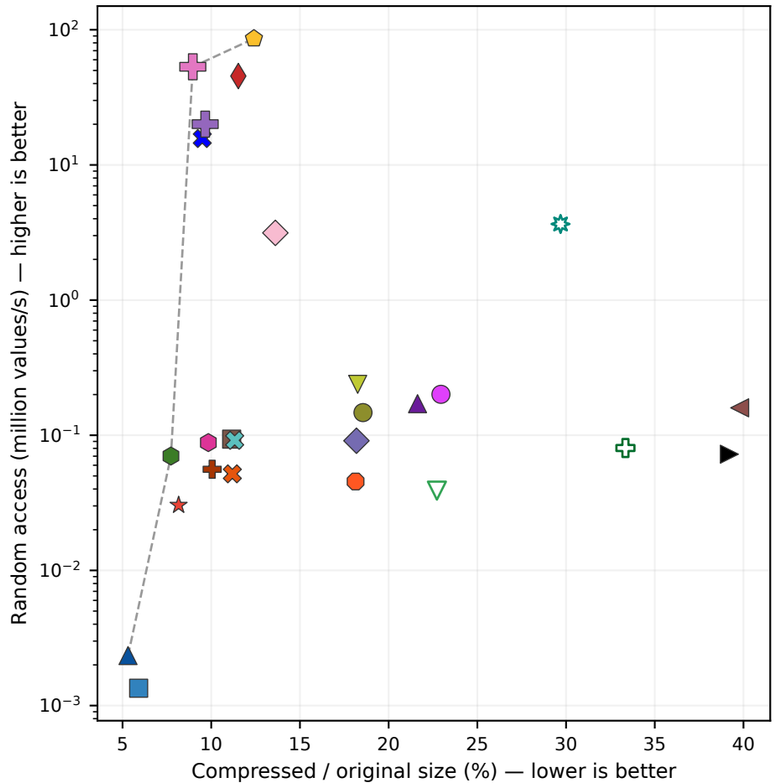
ALP
B*-GEF approx. [07d4e4a]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [07d4e4a]
SElf*
Snappy
TSXor
Xz
Zstd

GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

DU • optimized GEF 07d4e4a (uniform K=32,000) vs competitors

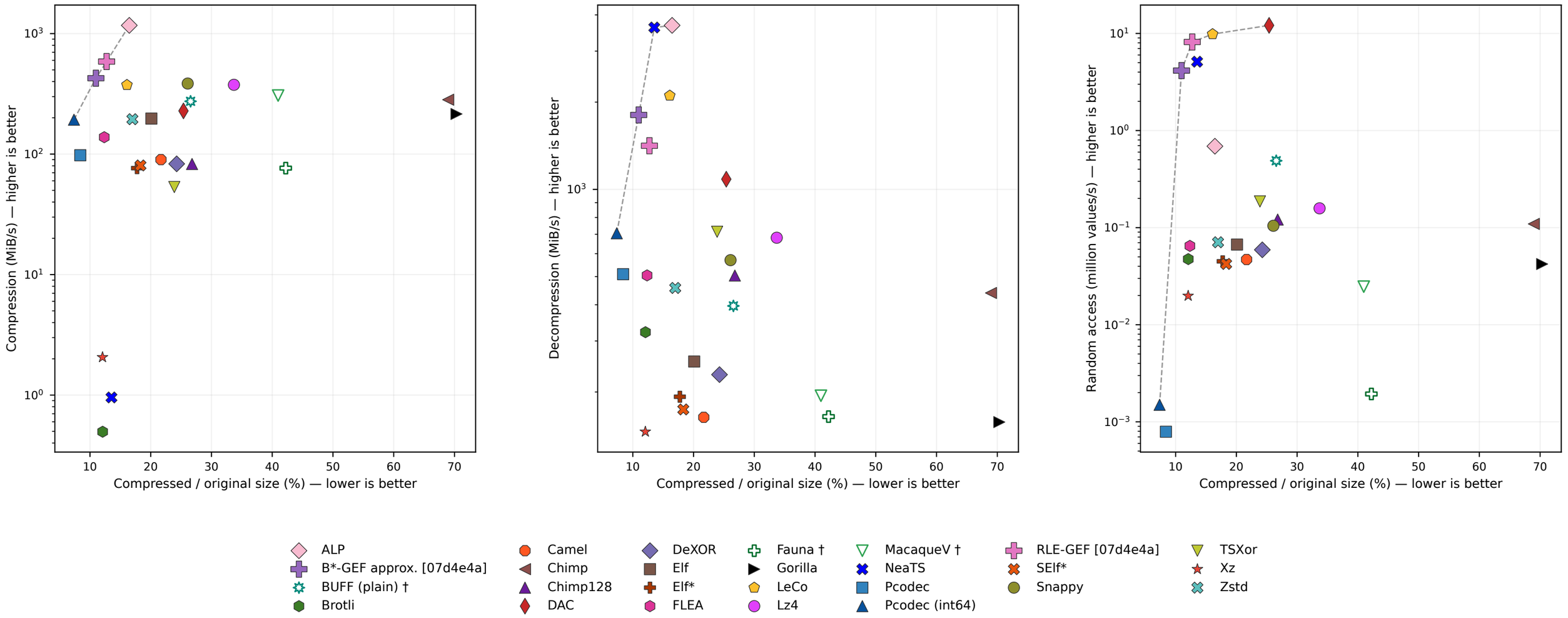




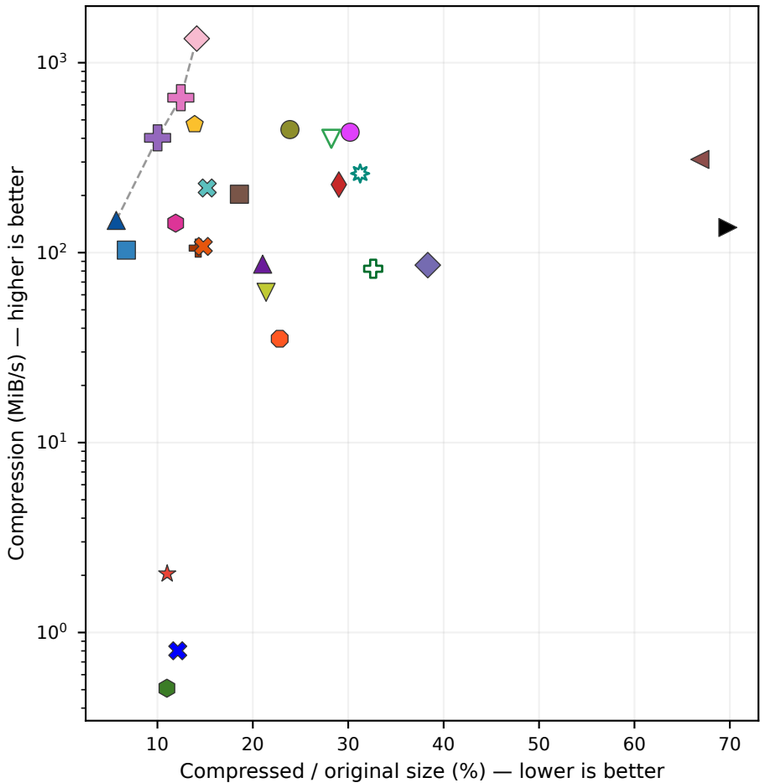


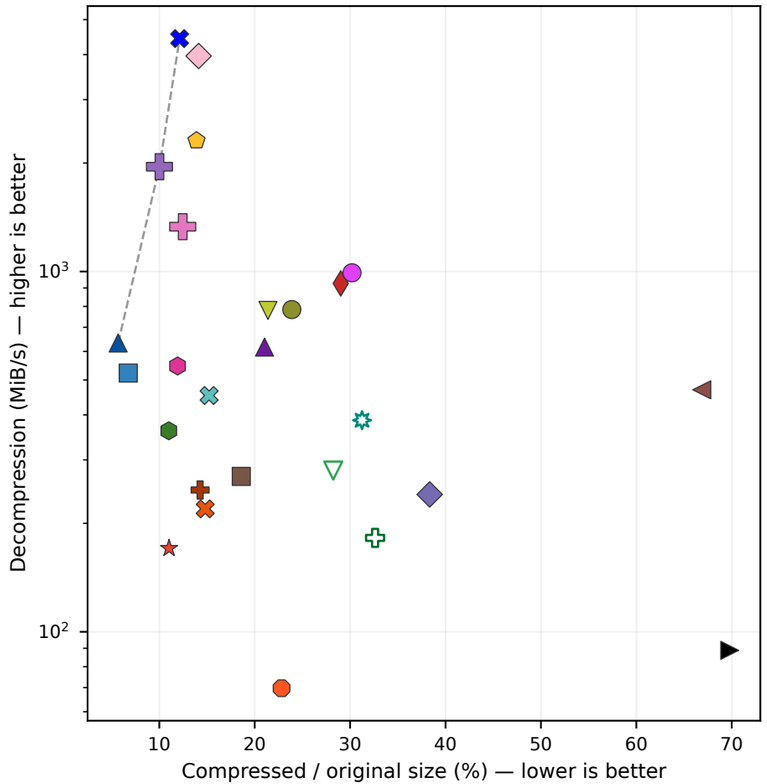
ALP
B*-GEF approx. [07d4e4a]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [07d4e4a]
SElf*
Snappy
TSXor
Xz
Zstd

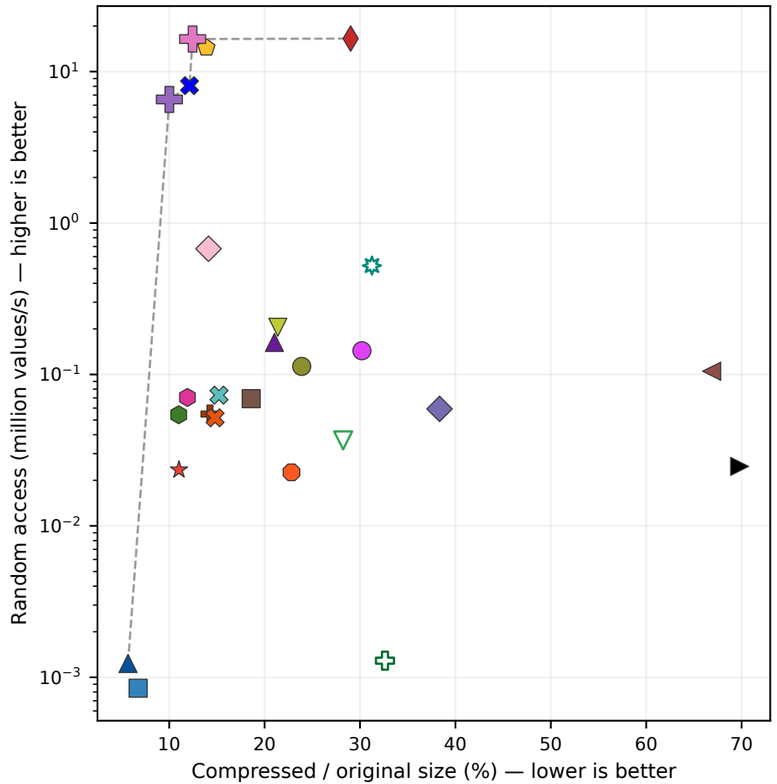
GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.


ECG • optimized GEF 07d4e4a (uniform K=32,000) vs competitors
Compression (MiB/s) — higher is better
Decompression (MiB/s) — higher is better
Random access (million values/s) — higher is better
Compressed / original size (%) — lower is better
ALP
B*-GEF approx. [07d4e4a]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [07d4e4a]
SElf*
Snappy
TSXor
Xz
Zstd
GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

GE • optimized GEF 07d4e4a (uniform K=32,000) vs competitors



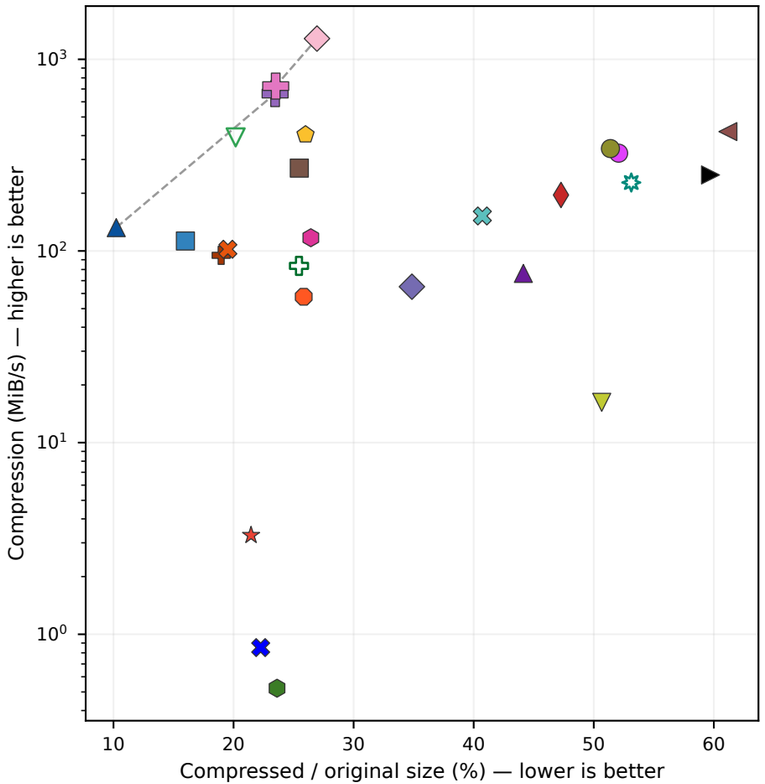


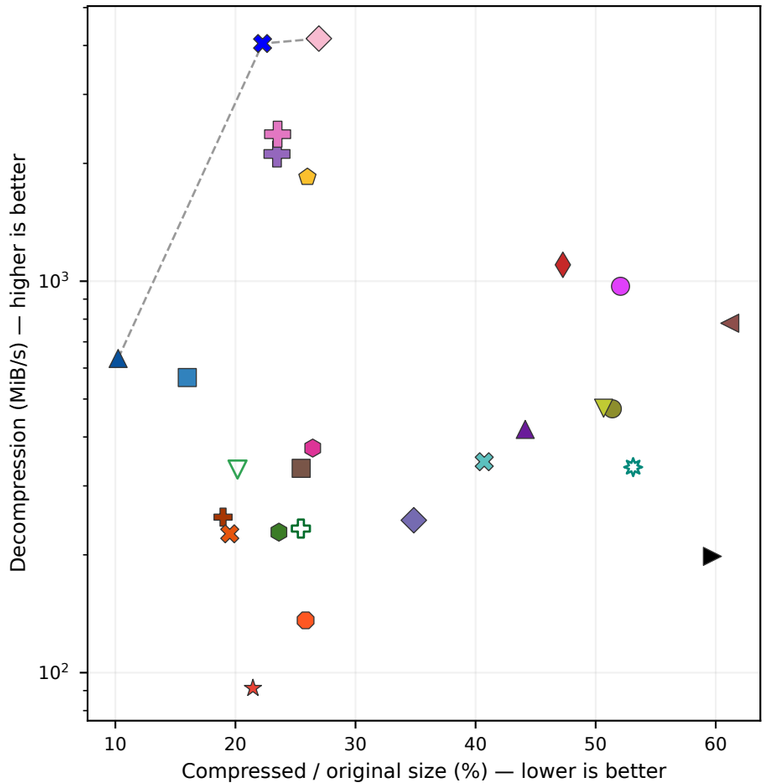


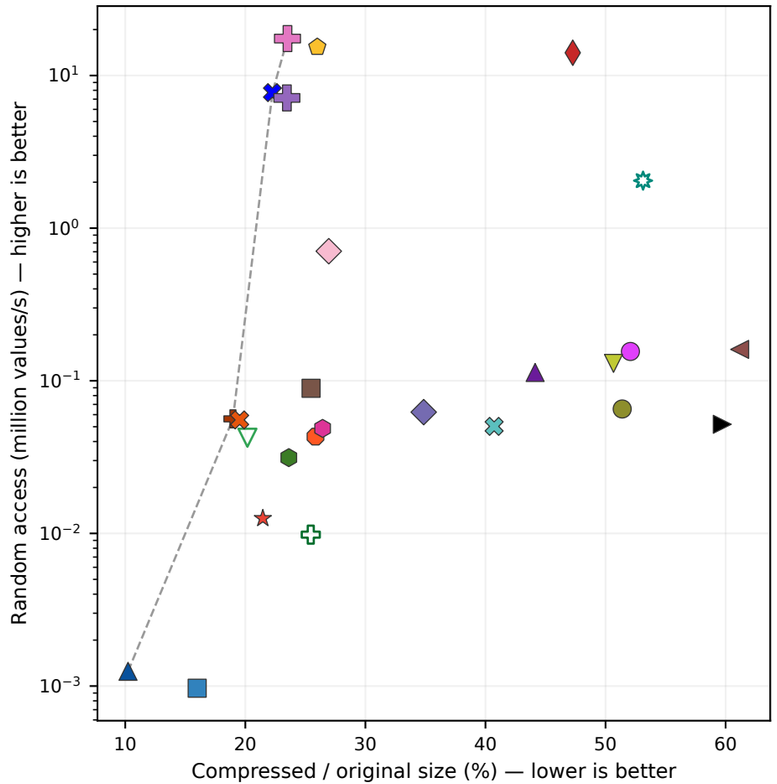
ALP
B*-GEF approx. [07d4e4a]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [07d4e4a]
SElf*
Snappy
TSXor
Xz
Zstd

GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

LAT • optimized GEF 07d4e4a (uniform K=32,000) vs competitors



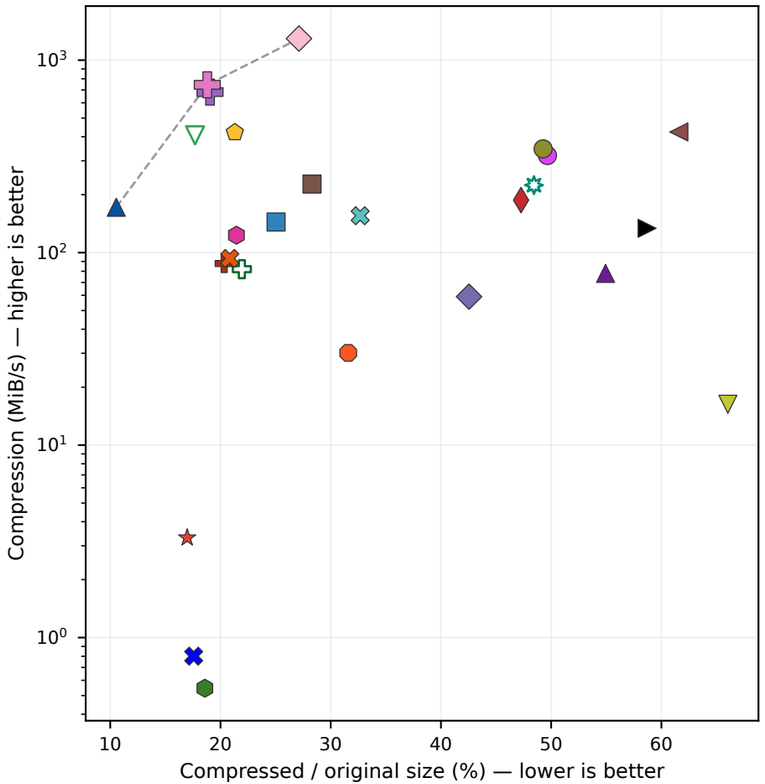


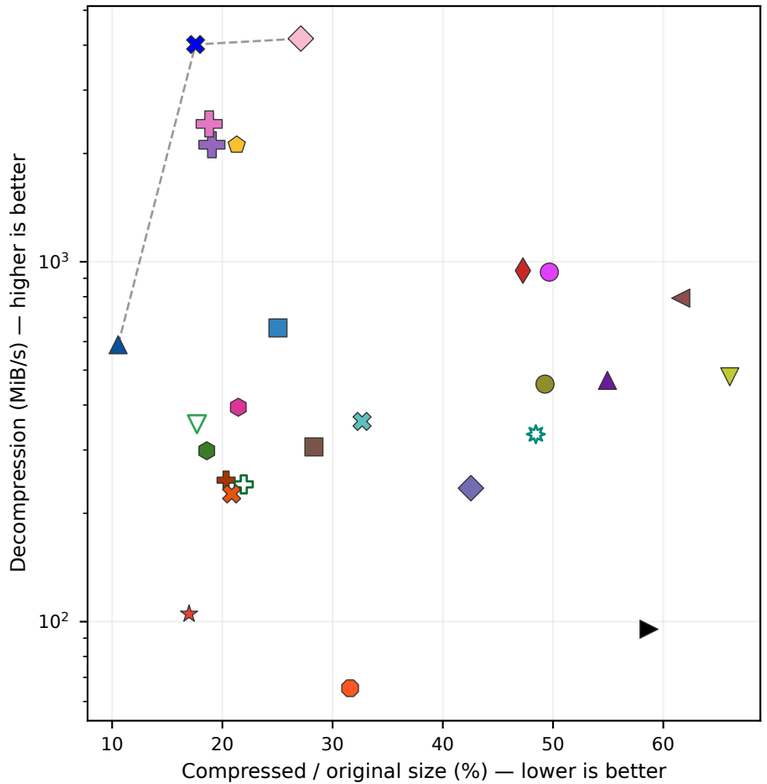


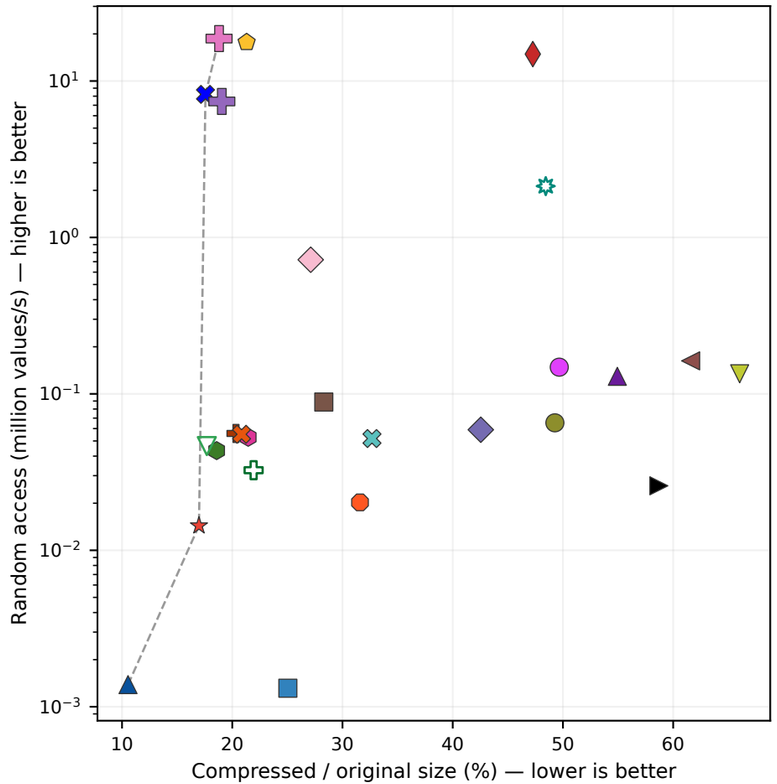
ALP
B*-GEF approx. [07d4e4a]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [07d4e4a]
SElf*
Snappy
TSXor
Xz
Zstd

GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

LON • optimized GEF 07d4e4a (uniform K=32,000) vs competitors



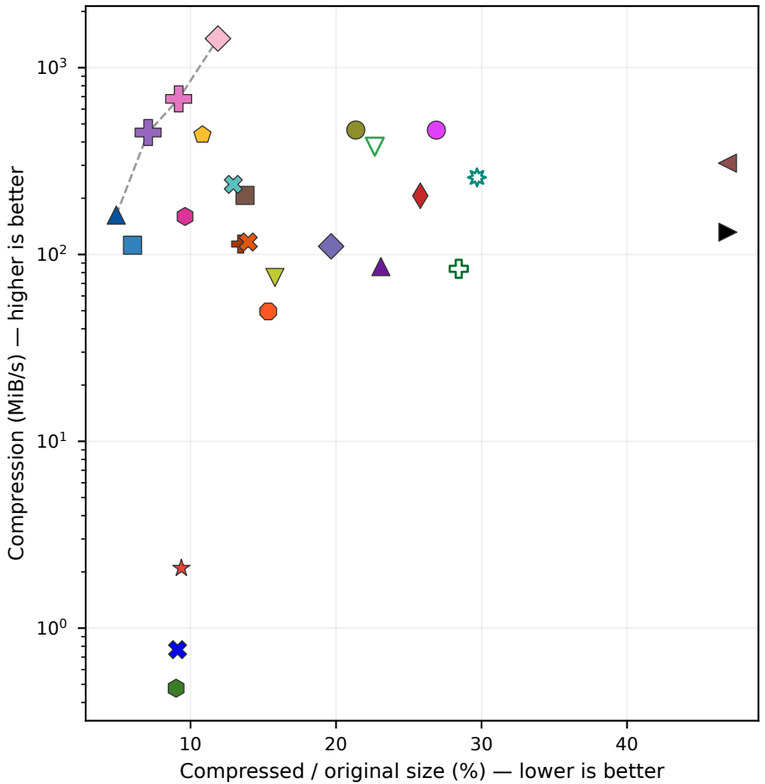


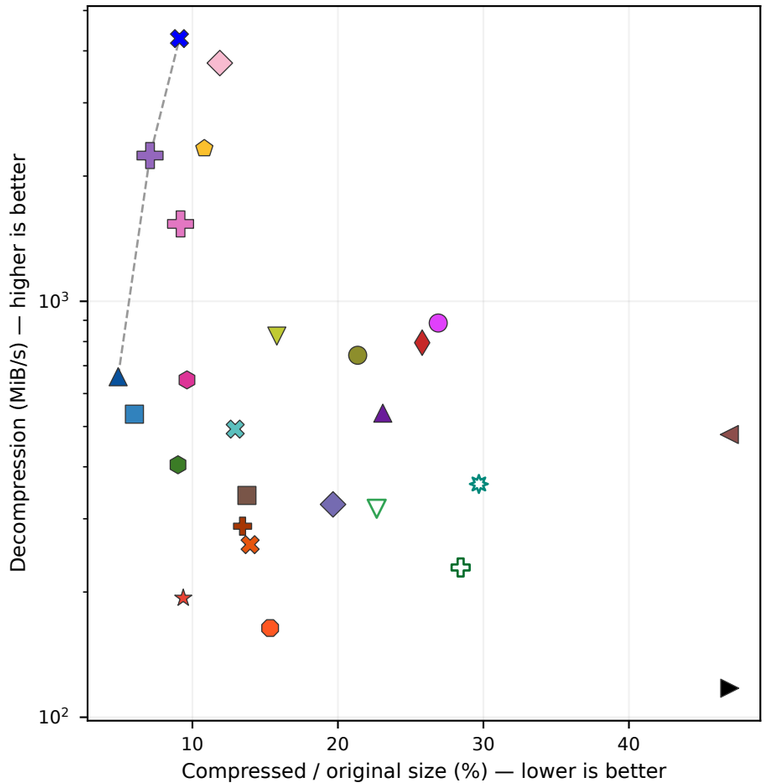


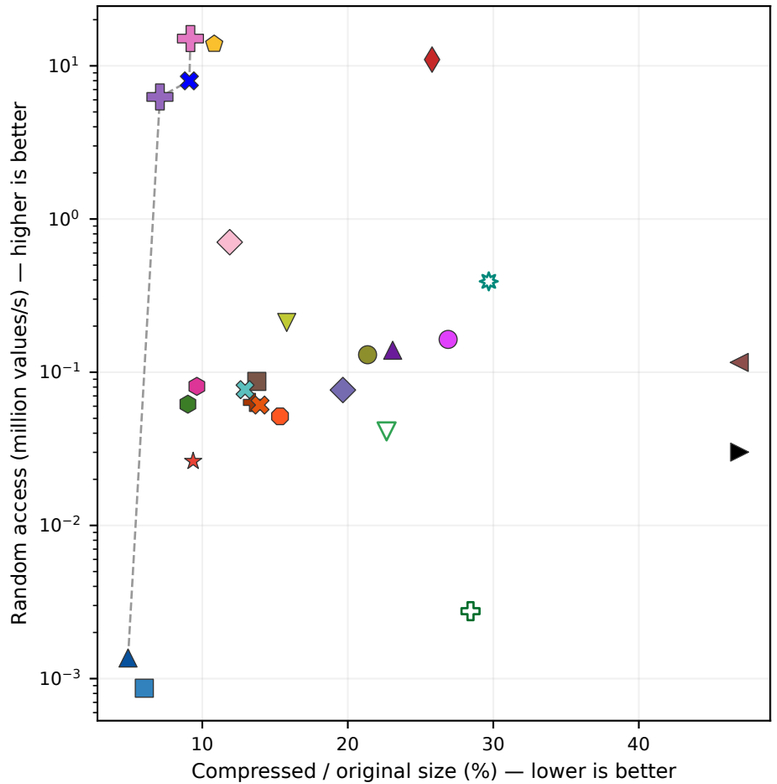
ALP
B*-GEF approx. [07d4e4a]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [07d4e4a]
SElf*
Snappy
TSXor
Xz
Zstd

GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

UK • optimized GEF 07d4e4a (uniform K=32,000) vs competitors

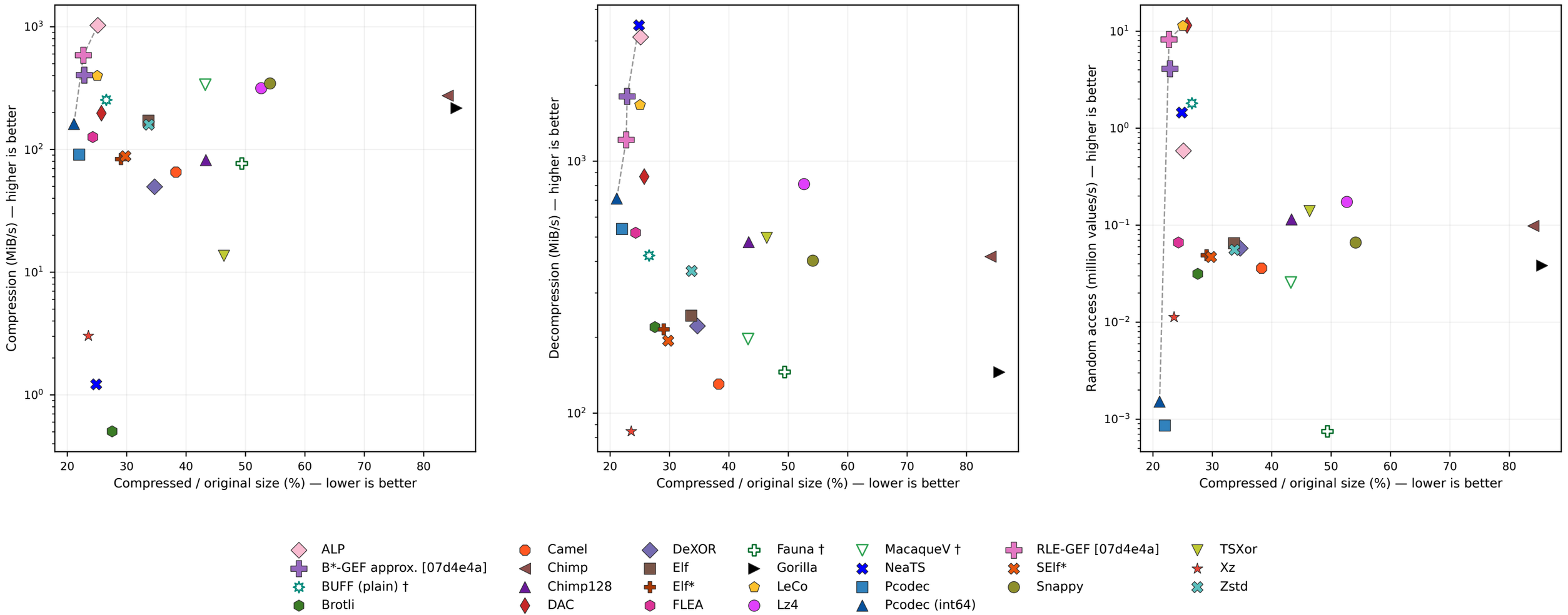






ALP
B*-GEF approx. [07d4e4a]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [07d4e4a]
SElf*
Snappy
TSXor
Xz
Zstd

GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.



GEF 07d4e4a: two independent full runs, three repetitions/run; mean of per-run median metrics.
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.